Solution to the Daily Limit

Based on the problem provided, we are tasked with first finding the function $g(x)$ and then evaluating a limit at infinity. The given expressions are:

$$g(x) = \lim_{r \to 0} \left((x+1)^{r+1} - x^{r+1}\right)^{\frac{1}{r}}$$

$$\text{Find: } \lim_{x \to \infty} \frac{g(x)}{x}$$

1. Evaluating $g(x)$

Let $L_r = ((x+1)^{r+1} - x^{r+1})^{\frac{1}{r}}$ such that $g(x) = \lim_{r \to 0} L_r$. We can evaluate this by taking the natural logarithm:

$$\ln g(x) = \lim_{r \to 0} \frac{\ln\left((x+1)^{r+1} - x^{r+1}\right)}{r}$$

As $r \to 0$, the argument of the logarithm approaches $(x+1)^1 - x^1 = 1$, making the numerator $\ln(1) = 0$. Since the denominator also approaches 0, we have an indeterminate form $\frac{0}{0}$. We apply L'Hôpital's Rule with respect to r:

$$\begin{aligned}
\ln g(x) &= \lim_{r \to 0} \frac{\frac{d}{dr}\left[\ln\left((x+1)^{r+1} - x^{r+1}\right)\right]}{\frac{d}{dr}[r]} \\
&= \lim_{r \to 0} \frac{\frac{(x+1)^{r+1}\ln(x+1) - x^{r+1}\ln(x)}{(x+1)^{r+1} - x^{r+1}}}{1}
\end{aligned}$$

Now, substituting $r = 0$ into the expression:

$$\begin{aligned}
\ln g(x) &= \frac{(x+1)^1 \ln(x+1) - x^1 \ln(x)}{(x+1)^1 - x^1} \\
&= (x+1)\ln(x+1) - x\ln(x)
\end{aligned}$$

Using logarithm properties, we can rewrite this as:

$$\ln g(x) = \ln\left((x+1)^{x+1}\right) - \ln\left(x^x\right) = \ln\left(\frac{(x+1)^{x+1}}{x^x}\right)$$

Exponentiating both sides yields $g(x)$:

$$g(x) = \frac{(x+1)^{x+1}}{x^x} = (x+1)\left(\frac{x+1}{x}\right)^x = (x+1)\left(1 + \frac{1}{x}\right)^x$$

2. Evaluating the Final Limit

We are asked to find $\lim_{x \to \infty} \frac{g(x)}{x}$. Substituting our simplified expression for $g(x)$:

$$\begin{aligned}
\lim_{x \to \infty} \frac{g(x)}{x} &= \lim_{x \to \infty} \frac{(x+1)\left(1 + \frac{1}{x}\right)^x}{x} \\
&= \lim_{x \to \infty} \left(\frac{x+1}{x}\right)\left(1 + \frac{1}{x}\right)^x \\
&= \lim_{x \to \infty} \left(1 + \frac{1}{x}\right) \cdot \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x
\end{aligned}$$

We evaluate the two limits separately. The first limit is trivial:

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)=1+0=1$$

The second limit is the standard definition of Euler's number e:

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}=e$$

Multiplying these results together:

$$\lim_{x\to\infty}\frac{g(x)}{x}=1\cdot e=e$$

Conclusion

After resolving the inner limit to find $g(x)$ and evaluating the outer limit as $x\to\infty$, the result converges precisely to Euler's number:

$$\lim_{x\to\infty}\frac{g(x)}{x}=e$$